

The EM Algorithm

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A Simple Measurement Problem

- Imagine that a factory produces the same component using two different machines.
- Let's say there is a common measurable quantity of the produce of both the machines, say the diameter of a component.
- Suppose:
 - ① Machine 1 tends to produce components around 20 mm.
 - ② Machine 2 tends to produce components around 25 mm.
- We collect measurements from the production line:

20.3, 24.8, 25.2, 19.7, 21.1, 24.5, 20.8, 25.4, ...

- But there is a problem.

We did not record which machine produced each component

What Would We Like to Know?

- We would like to recover information about the two machines.
- For example:

μ_1 = average output of Machine 1

μ_2 = average output of Machine 2

σ_1 = variability of Machine 1

σ_2 = variability of Machine 2

- We might also want to know:

π_1 = fraction of production from Machine 1

π_2 = fraction of production from Machine 2

- So from the measurements alone, we want to reconstruct the characteristics of the two machines.

Machine Information Were Available VS Labels Are Missing

- Suppose the production records had been maintained properly:

Measurement	20.3	24.8	25.2	19.7	21.1	24.5	20.8	25.4
Machine	1	2	2	1	1	2	1	2

- Then the problem would be easy. We could separate the measurements and calculate the mean and variance of each group.
- In reality, all we have is
20.3, 24.8, 25.2, 19.7, 21.1, 24.5, 20.8, 25.4, ...
- We don't know which machine produced which measurement. This missing information is the central difficulty.

Why Can't We Just Separate the Data?

- Looking at the data, we might say:
Values near 20 probably came from Machine 1, and values near 25 probably came from Machine 2.
- But consider $x = 22.8$. Which machine produced it?
- Or suppose the two machines have more variability. There is **no sharp boundary** separating the two populations.
- So assigning $x = 22.8$ completely to one machine may be unreasonable.

A Better Question

- Instead of asking
Which machine produced this observation?
- we ask
How likely is each machine to have produced this observation?

A Better Question

- For example, for $x = 22.8$, we might obtain

	Machine 1	Machine 2
$x = 22.8$	0.7	0.3

- We don't have to make a hard decision. We can say:

70% associated with Machine 1, 30% associated with Machine 2.

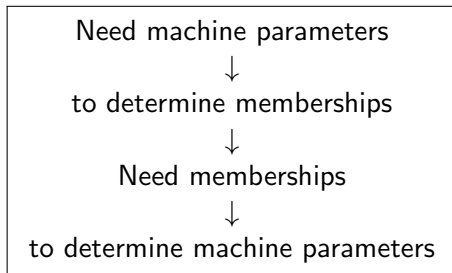
- How are these numbers of any help?
- These numbers will become **weights** when estimating the machine parameters.
- All well and good still we only have raw data, how do we assign any likelihood this data?
- this will need us to know more about the machines, isn't it !!

Now We Have a Circular Problem

- It is an interesting and peculiar difficulty.
- To decide how likely each machine is to have produced x , we need to know the characteristics of the machines.
- For example, we need to know

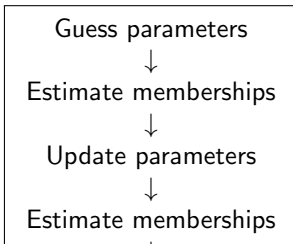
$$\mu_1, \sigma_1, \mu_2, \sigma_2.$$

- But to estimate those characteristics, we need to know which measurements came from each machine! So we have a circular problem:



Breaking the Circle

- This is exactly the situation where EM becomes useful.
- We can break this circularity by **alternating between the two tasks**.
- Start with some rough guesses for the machine parameters. Then:
 - 1 Use our current guesses to estimate how strongly each observation belongs to each machine.
 - 2 Use these estimated memberships to improve the machine parameters.
- Then use the improved parameters to estimate the memberships again. So:



A Very Simple Analogy

- Think of two teachers trying to determine which of two courses a set of students most likely belongs to.
- Initially, they have rough ideas about the typical students in each course.
- They then:
 - 1 **Estimate:** Based on the current descriptions, how strongly does each student resemble each course?
 - 2 **Update:** Based on these assignments, revise the description of each course.
 - 3 **Repeat.**
- The descriptions become better, which makes the assignments better, which makes the descriptions better again.
- The intuition behind EM:

better assignments	$\longleftrightarrow$	better parameters
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Where Does the Name EM Come From?

- The two alternating steps have names.
 - ① **E – Expectation:** Estimate the **expected/soft membership** of each observation.

Which source is responsible, and by how much?

- ② **M – Maximization:** Using those memberships, find the parameters that best describe the data.

What parameters best fit these weighted observations?

E-step $\longrightarrow$ M-step $\longrightarrow$ E-step $\longrightarrow \dots$

Set up the model

- We assume that every observation comes from one of two Gaussian populations:

$$X \sim \begin{cases} \mathcal{N}(\mu_1, \sigma_1^2), & \text{with probability } \pi_1, \\ \mathcal{N}(\mu_2, \sigma_2^2), & \text{with probability } \pi_2. \end{cases}$$

- where

$$\pi_1 + \pi_2 = 1.$$

- The unknown parameters are

$$\theta = (\mu_1, \sigma_1^2, \mu_2, \sigma_2^2, \pi_1, \pi_2).$$

- **Goal:** Estimate these parameters from observations

$$x_1, x_2, \dots, x_N$$

when the source population of each observation is unknown.

Why choose a Gaussian Model

- The Gaussian distribution is a natural first model for measurements that fluctuate around a typical value.

- Its density:

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

μ = center σ = spread.

- So instead of describing a machine using thousands of measurements, we can describe it using just : μ σ^2
- When many approximately independent small effects add together, their combined effect is often approximately Gaussian.
- So Gaussian models arise naturally in measurement and engineering problems.
- A uniform distribution would say that all values in some interval are equally likely. For example, $x \in [18, 22]$.
- But measurements near 20 would normally be more common than measurements near 18 or 22.

The Gaussian Density

- For population k , the Gaussian density is

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right).$$

- If one population is much more common than the other, that should also affect our assignment. Therefore we multiply by the population proportion:

$$\pi_k f_k(x_i)$$

- The two populations therefore have densities

$$f_1(x) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(x - \mu_1)^2}{2\sigma_1^2}\right),$$

$$f_2(x) = \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(x - \mu_2)^2}{2\sigma_2^2}\right).$$

The Responsibilities

- For two populations, define

$$w_{i1} = \pi_1 f_1(x_i),$$

$$w_{i2} = \pi_2 f_2(x_i),$$

- These are not yet probabilities. Normalize them:

$$r_{i1} = \frac{w_{i1}}{w_{i1} + w_{i2}}, \quad r_{i2} = \frac{w_{i2}}{w_{i1} + w_{i2}}.$$

- Equivalently,

$$r_{i1} = \frac{\pi_1 f_1(x_i)}{\pi_1 f_1(x_i) + \pi_2 f_2(x_i)}, \quad r_{i2} = \frac{\pi_2 f_2(x_i)}{\pi_1 f_1(x_i) + \pi_2 f_2(x_i)}$$

- These are the responsibilities. They satisfy

$$r_{i1} + r_{i2} = 1$$

What does a responsibility mean?

- Suppose for one observation we obtain $r_{i1} = 0.8$ and $r_{i2} = 0.2$.
- We interpret this as:
The current model considers this observation to be 80% associated with population 1 and 20% associated with population 2.
- Importantly, we do not actually assign it to population 1. We retain both numbers.
- This is the key difference between EM and a simple clustering procedure based on hard assignments.

E-Step: Responsibilities

- For each observation x_i , we ask:

How likely is x_i to have come from each Gaussian?

- For Gaussian k , with density function f_k , the responsibility of component k for observation x_i is

$$r_{ik} = \frac{\pi_k f_k(x_i)}{\sum_j \pi_j f_j(x_i)}$$

where

$$\sum_k r_{ik} = 1.$$

- r_{ik} is the **soft membership** of x_i in component k .

M-Step: Update the Parameters

- Once the responsibilities are known, use them as weights to update the parameters.
- Define the effective number of observations in component k :

$$N_k = \sum_{i=1}^N r_{ik}.$$

- The new mean is the weighted mean:

$$\mu_k = \frac{\sum_{i=1}^N r_{ik} x_i}{N_k}$$

- The new variance is the weighted variance:

$$\sigma_k^2 = \frac{\sum_{i=1}^N r_{ik} (x_i - \mu_k)^2}{N_k}$$

- and the new mixing fraction is

$$\pi_k = \frac{N_k}{N}.$$

The Complete EM Algorithm

Initialize

$$\mu_k, \quad \sigma_k^2, \quad \pi_k.$$

Repeat until convergence:

- 1 **E-step:** Compute responsibilities

$$r_{ik} = \frac{\pi_k f_k(x_i)}{\sum_j \pi_j f_j(x_i)}.$$

- 2 **M-step:** Update

$$\begin{aligned}\mu_k &= \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}, \\ \sigma_k^2 &= \frac{\sum_i r_{ik} (x_i - \mu_k)^2}{\sum_i r_{ik}}, \\ \pi_k &= \frac{\sum_i r_{ik}}{N}.\end{aligned}$$

E-step $\longrightarrow$ M-step $\longrightarrow$ E-step $\longrightarrow \dots$

Problem: Fit a Two-Gaussian Mixture

A factory produces components using two machines. The machine responsible for each measurement is unknown.

The measured diameters are:

18.9	19.4	19.8	20.1	20.3	20.7
21.0	21.4	21.8	22.1	22.5	22.8
23.2	23.5	23.9	24.2	24.5	24.8
25.1	25.4	25.8	26.1	26.5	26.9

Start with

$$\mu_1 = 20, \quad \mu_2 = 26,$$

$$\sigma_1^2 = \sigma_2^2 = 4, \quad \pi_1 = \pi_2 = 0.5.$$

Task: Apply the EM algorithm to estimate